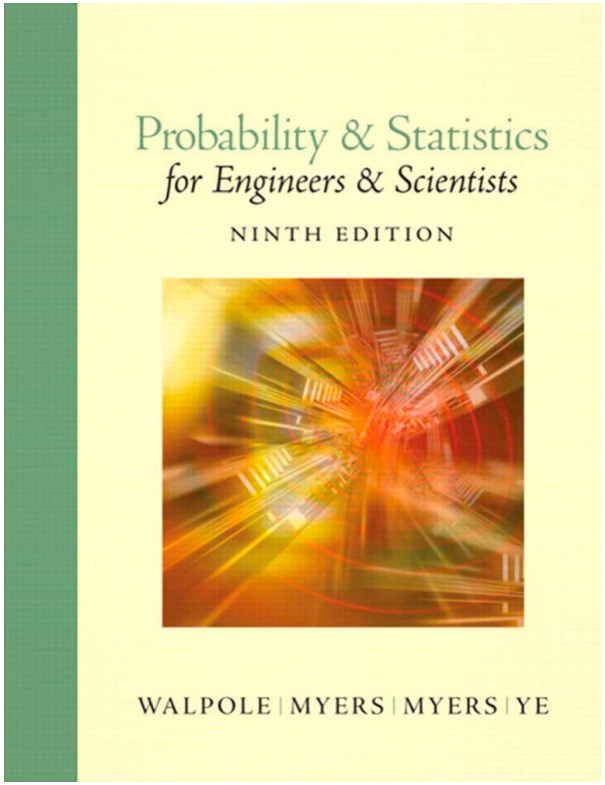
Regression

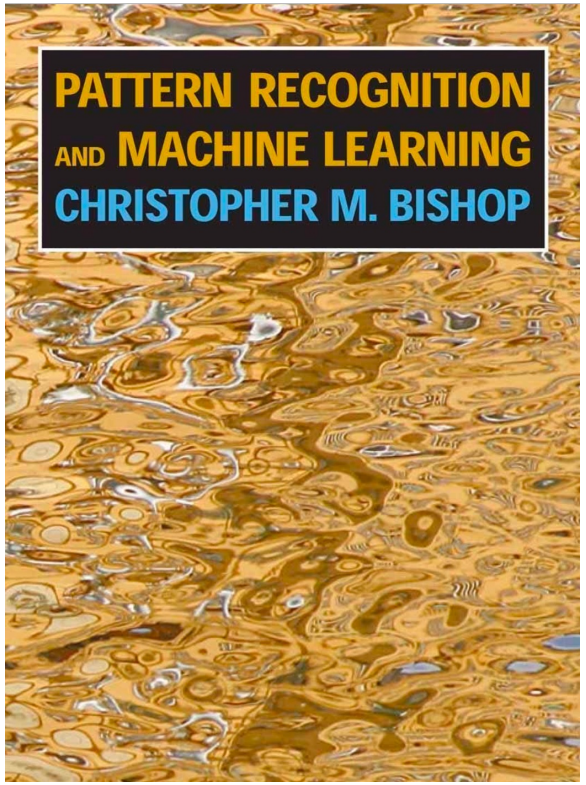
Walpole “Chapter 11”

Topic

- Introduction
- Simple Linear Regression Model →
  - *The Fitted Regression Line*

Fundamental concept in data science and machine learning.





Introduction

Example:

In an industrial process, it is known that the likelihood of **tar** content in a chemical solution within a pipe is related to the temperature at the *pipe's inlet*.

- The tar content in the chemical solution ➔ dependent variable or response (Y)
- Temperature ➔ natural independent variable or regressor (regressor variable ➔ x)
- Most likely, in reality, at the same temperature (e.g., 130°C), the amount of tar in the chemical solution will not be the same.

- Prediction method → "estimation of chemical solution content for a given temperature."
- The possible relationship between the response (Y) and the regressor x is a linear relationship as follows:

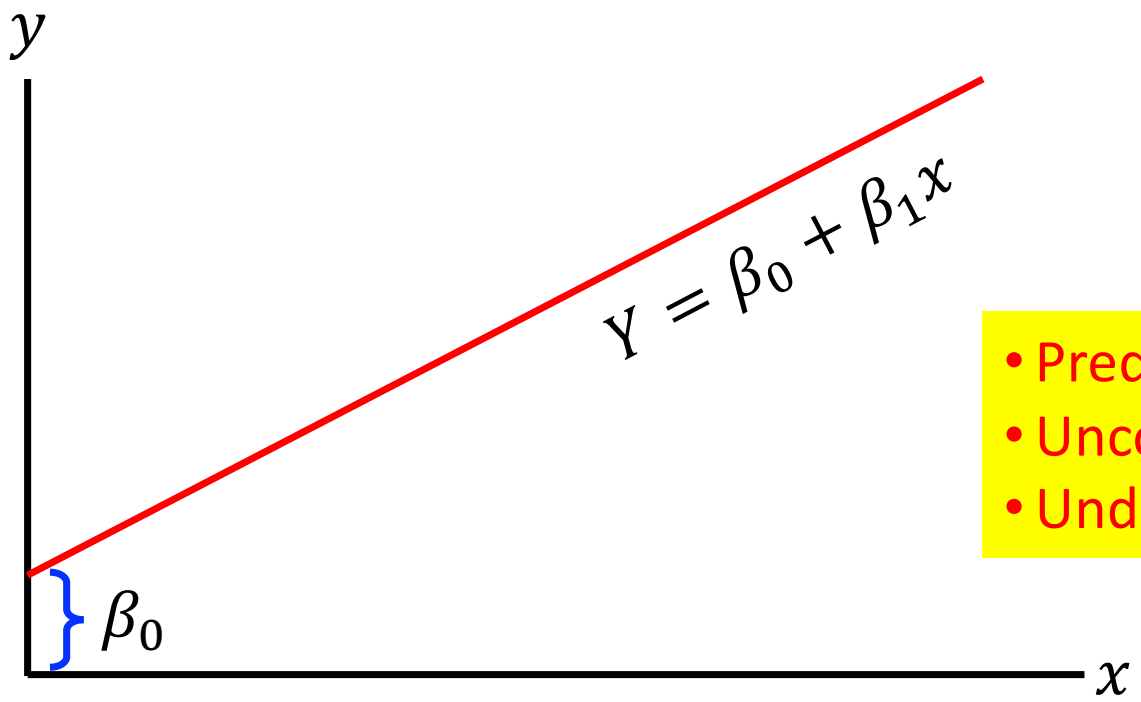
$$Y = \beta_0 + \beta_1 x$$

β_0 is the intercept (intersection with the Y-axis), and β_1 is the slope/gradient.

Possible relationships between two variables $(Y \text{ and } x)$:

1. Deterministic → no random component
2. Non-deterministic (a certain x value does not always give the same Y value) → probabilistic

- The concept of *regression* analysis is related to finding the best relationship between Y and x.



- Predict
- Uncover Pattern
- Understand Data

- If the regression model involves more than one regressor, then it is called multiple regression.

$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2$$

Simple Linear Regression (SLR) Model

$$Y = \beta_0 + \beta_1 x \quad \Rightarrow \quad \textit{Simple Linear Regression}$$

$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 \quad \Rightarrow \quad \textit{Multiple Regression}$$

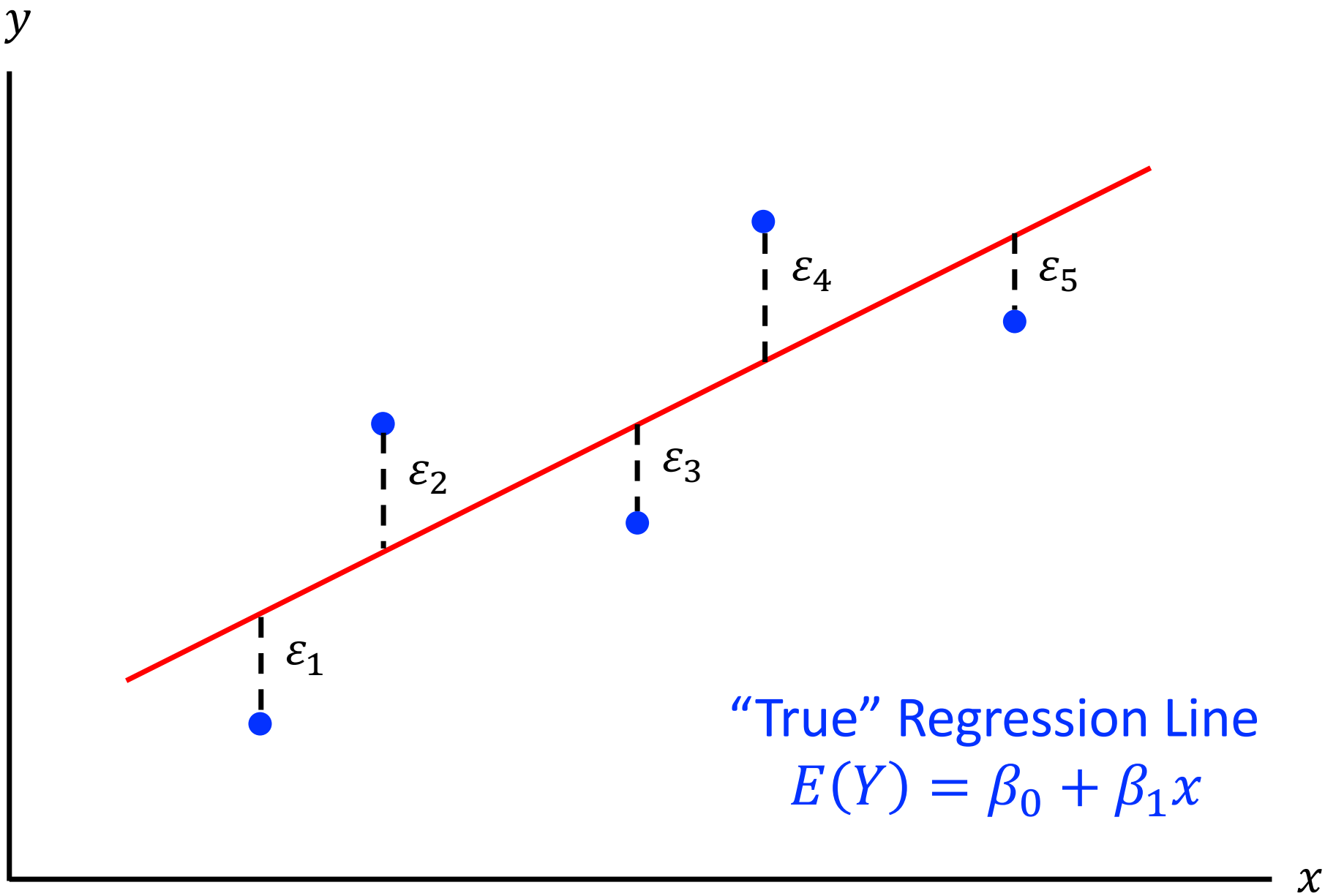
- Random sample (size n) →$\{(x_i, y_i);\ i = 1, 2, \dots, n\}$.
- If *extra samples* are taken using the same value of x_i, then the value of y_i will vary → the value y_i in the successive pair (x_i, y_i) is the value of a random variable Y_i
- There is a *random component* in the SLR equation that realizes both variables → this random component is *unseen*, *unmeasurable*, and *possibly not understood* by scientists/engineers.

- The model $Y = \beta_0 + \beta x \rightarrow$ *an approximation* of something more complex or not precisely known.
- The response Y is related to the independent variable x as follows:

$$Y = \beta_0 + \beta_1 x + \epsilon$$

ϵ is a random variable assumed to be distributed such that:

$E(\epsilon) = 0 \; and \; Var(\epsilon) = \sigma^2$. The value σ^2 represents the *error variance*.


y
ε_1
ε_2
ε_3
ε_4
ε_5
"True" Regression Line
$E(Y) = \beta_0 + \beta_1 x$
x

The Fitted Regression Line

- The *estimated regression coefficient* is defined as b_0 as the estimate for β_0 , and b_1 as the estimate for β_1. Then the *estimated or fitted regression line* is given by:

$$\hat{y} = b_0 + b_1 x$$

$\hat{y}$ → *predicted* or *fitted* value.

- From the values of $\hat{y}$ → *Fitted line* → an estimate of the *"true regression line"*
- Ideally → the *fitted line* becomes closer to the *true regression line* as the amount of data obtained increases.

Solids Reduction, x (%)	Oxygen Demand Reduction, y (%)	Solids Reduction, x (%)	Oxygen Demand Reduction, y (%)
3	5	36	34
7	11	37	36
11	21	38	38
15	16	39	37
18	16	39	36
27	28	39	45
29	27	40	39
30	25	41	41
30	35	42	40
31	30	42	44
31	40	43	37
32	32	44	44
33	34	45	46
33	32	46	46
34	34	47	49
36	37	50	51
36	38		

same regression but different value Y. In reality, this is example same regression

The Fitted Regression Line

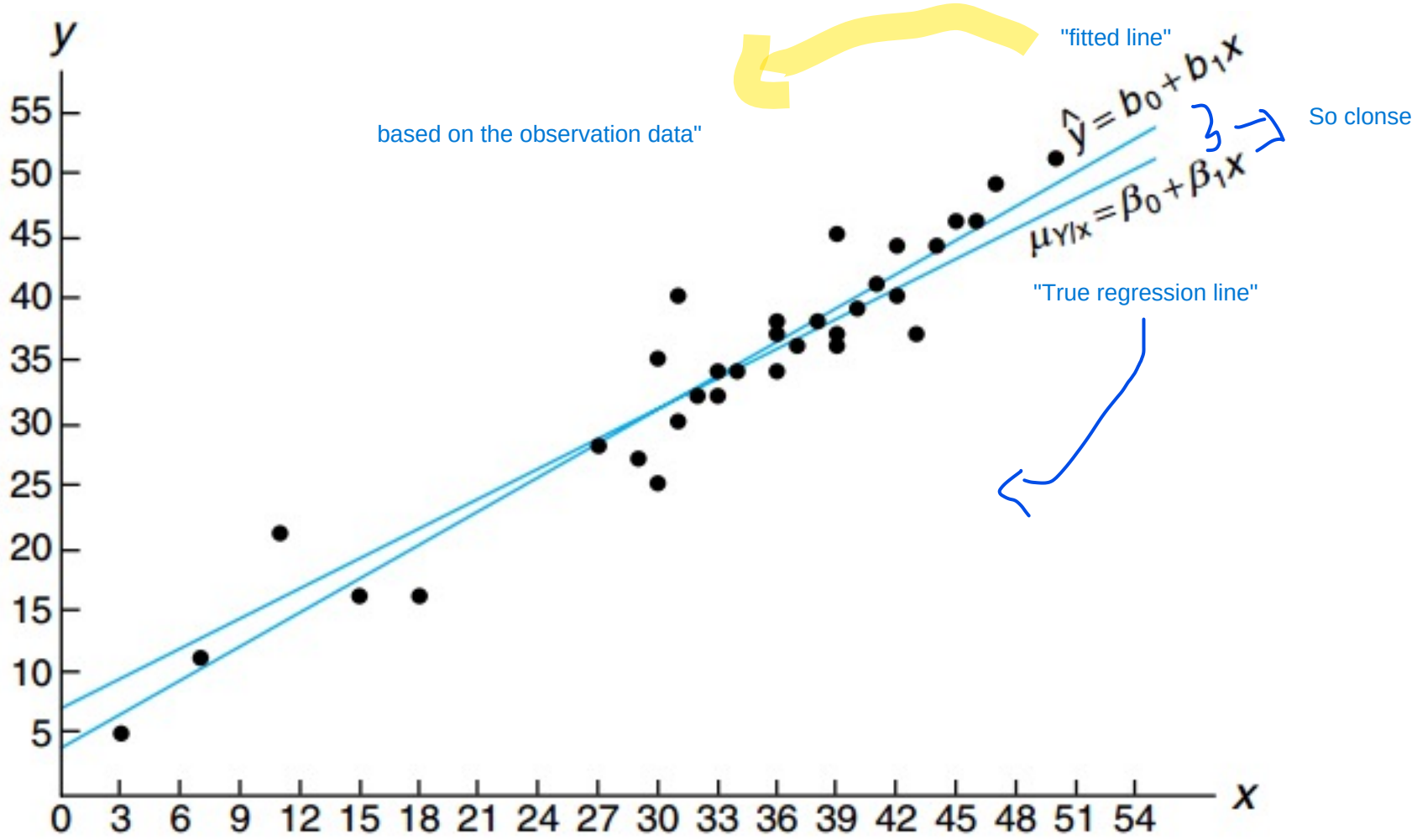


Figure 11.3 of Walpole, et al (2012), Probabilty & Statistics For Engineers & Scientists: Example of a scatter diagram (see Table 11.1) with the fitted regression line and a hypothetical true regression line.

Model Assumption on True Regression Line

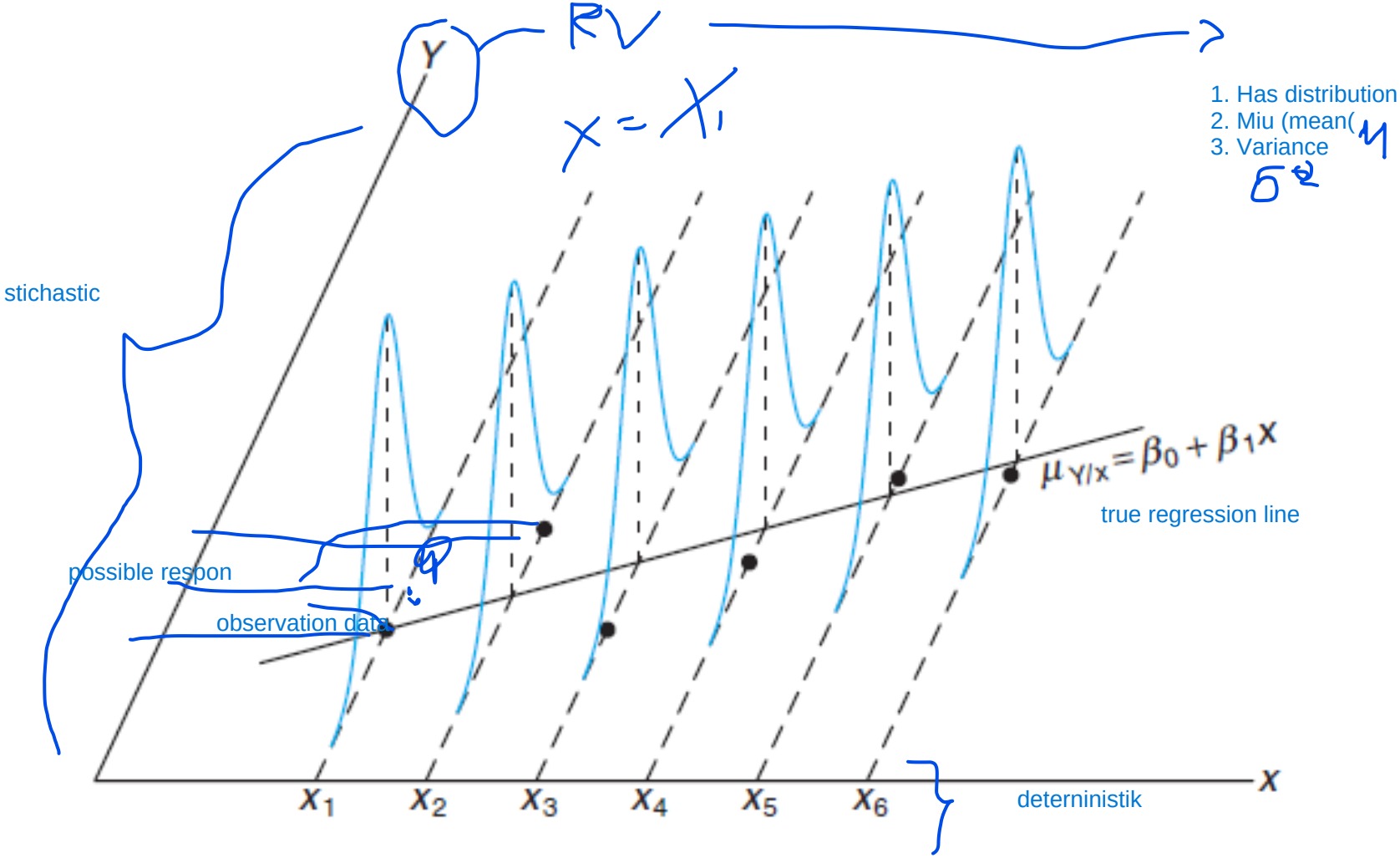


Figure 11.4 of Walpole, et al (2012), Probabilty & Statistics For Engineers & Scientists: Individual Observation around the True Regression Line.

- The line in the figure: *True Regression Line*
- The points scattered around the line ➔ observed (x, y) pair.
- Each *observation point* has its own normal curve, with the center of the normal curve located on the line.
- $E[Y] = \beta_0 + \beta_1 x \Rightarrow$ thus the true regression line: goes through *the mean of the response*.
- Each observation point → *on the distribution around the mean* → has the same variance, denoted by σ^2.

true regresion

$$y_i - E(Y_i) = y_i - (\beta_0 + \beta_1 x_1) = \epsilon_i$$

its observation

Error and Residual Concept

- Defintion: Given *a set of regression data*: $\{(xi, yi);\ i = 1,2, \dots, n\}$ and *a fitted model* $\hat{y}_i = \beta_0 + \beta_1 x_i$, *the* i^{th} *residual* e_i is given by:

$$e_i = y_i - \hat{y}_i, \qquad i = 1,2, \dots, n$$

- Another relationship is the following:

ε : difference between obs. and true

$$y_i = b_0 + b_1 x_i + e_i$$

e : difference between obs and fitted -> we cannot measure this

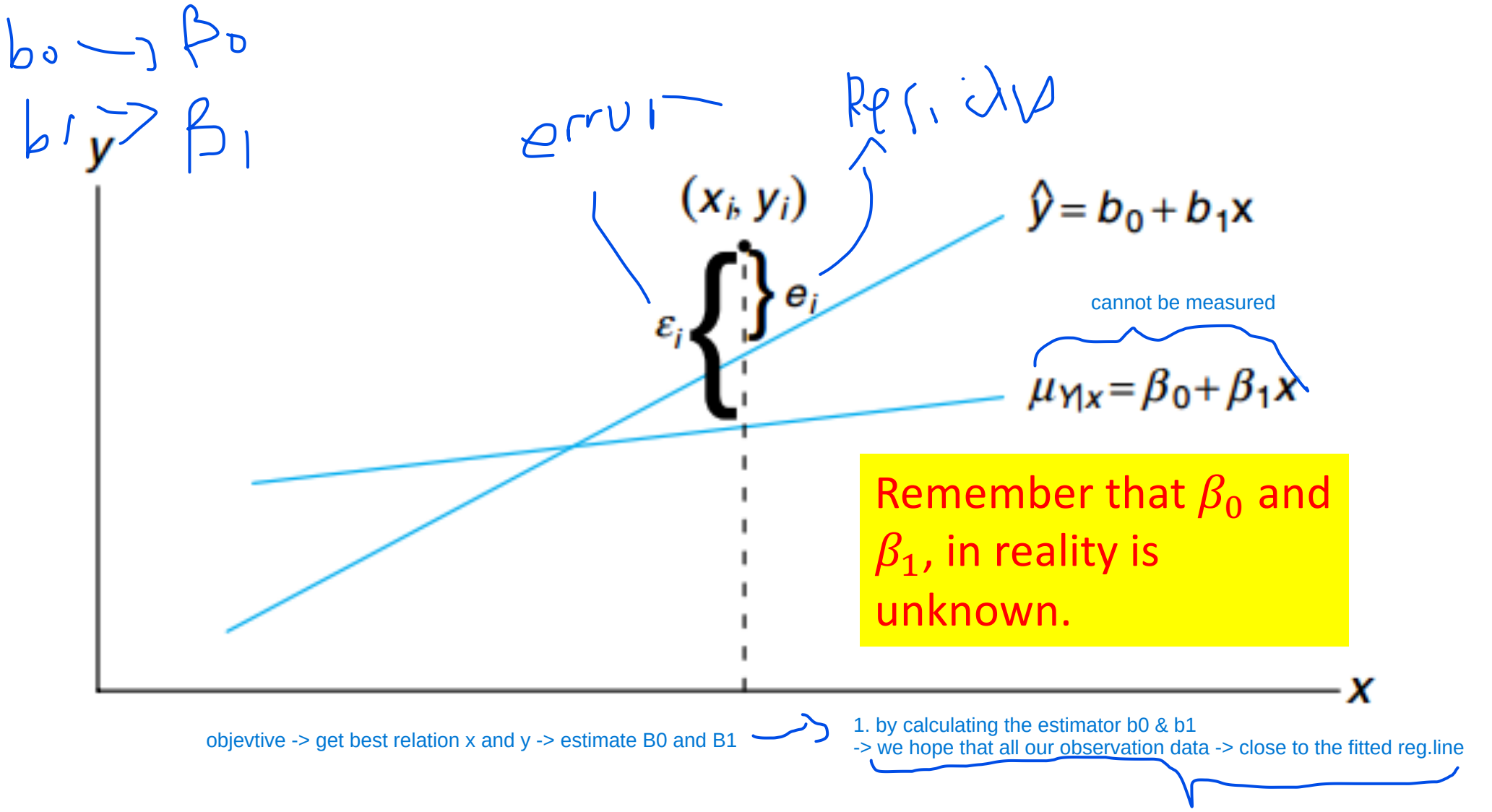


Figure 11.5 of Walpole, et al (2012), Probabilty & Statistics For Engineers & Scientists: Comparing random error ϵ_i dengan residual e_i.

MSE : ^2
MAE (min Absolut Error) ^1

The Method of Least Squares

Mencari selisih error dalam hal ini residu, dengan meminimalkan

- **Objective:** To find b_0 dan b_1, estimate of β_0 dan β_1, ➔ so that the sum of the squares of the residuals is a minimum.
- The sum of the squares of the residuals ➔ **SSE** ➔ *"This can be somewhat misleading, as in practice it is the residuals that are minimized, not the random error itself."*

$y_i = b_0 + b_1 x$

$$SSE = \sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 = \sum_{i=1}^{n} (y_i - b_0 - b_1 x_i)^2$$

Residu

Estimating the Regression Coefficient

- Differentiate SSE with respect to b_0 dan b_1:

$$\frac{\partial(SSE)}{\partial b_0} = -2\sum_{i=1}^{n}(y_i - b_0 - b_1 x_i) \qquad \frac{\partial(SSE)}{\partial b_1} = -2\sum_{i=1}^{n}(y_i - b_0 - b_1 x_i)x_i.$$

- Then set the result of differentiation equal to zero ➔ normal equations

$$nb_0 + b_1\sum_{i=1}^{n} x_i = \sum_{i=1}^{n} y_i, \quad \textbf{(1)}$$

$$b_0\sum_{i=1}^{n} x_i + b_1\sum_{i=1}^{n} x_i^2 = \sum_{i=1}^{n} x_i y_i, \quad \textbf{(2)}$$

- Next derivation: Multiply (2) by n and multiply (1) by:

$$\sum_{i=1}^{n} x_i \quad \text{then eliminate } b_0.$$

Estimating Regression Coefficient

Given the sample $\{(xi, y_i);\ i = 1, 2, \ldots, n\}$, the *Least Squares* estimates b_0 and b_1 of the regression coefficients β_0 and β_1 are computed from the formulas:

$$b_1 = \frac{n\sum_{i=1}^{n} x_i y_i - \left(\sum_{i=1}^{n} x_i\right)\left(\sum_{i=1}^{n} y_i\right)}{n\sum_{i=1}^{n} x_i^2 - \left(\sum_{i=1}^{n} x_i\right)^2} = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

$$b_0 = \frac{\sum_{i=1}^{n} y_i - b_1 \sum_{i=1}^{n} x_i}{n} = \bar{y} - b_1\bar{x}$$

Example Data

Solids Reduction, x (%)	Oxygen Demand Reduction, y (%)	Solids Reduction, x (%)	Oxygen Demand Reduction, y (%)
3	5	36	34
7	11	37	36
11	21	38	38
15	16	39	37
18	16	39	36
27	28	39	45
29	27	40	39
30	25	41	41
30	35	42	40
31	30	42	44
31	40	43	37
32	32	44	44
33	34	45	46
33	32	46	46
34	34	47	49
36	37	50	51
36	38		

Estimate the regression line for the pollution data

$$\sum_{i=1}^{33} x_i = 1104, \sum_{i=1}^{33} y_i = 1124, \sum_{i=1}^{33} x_i y_i = 41,335,$$

$$\sum_{i=1}^{33} x_i^2 = 41,086$$

$$b_1 = \frac{(33)(41{,}355) - (1104)(1124)}{(33)(41{,}086) - (1104)^2} = 0.903643$$

$$b_0 = \frac{1124 - (0.903643)(1104)}{33} = 3.829633.$$

$$\hat{y} = 3.8296 + 0.9036x$$

Example 1

The grades of a class of 9 students on a midterm report (x) and on the final examination (y) are as follows:

x	77	55	71	72	85	94	96	99	67
y	82	66	78	34	47	85	99	99	68

a) Estimate the linear regression line.
b) Estimate the final examination grade of a student who received a grade of 85 on the midterm report.

Example 2

The following data are the selling prices *z* of a certain make and model of used car *w* years old. Fit a curve of the form $\mu_{z|w} = \gamma\delta^w$ by means of the nonlinear sample regression equation $\hat{z} = cd^w$. [Hint: Write $\ln \hat{z} = \ln c + (\ln d)w = b_0 + b_1 w$]

W(years)	Z(dollars)
1	6350
2	5695
2	5750
3	5395
5	4985
5	4895

Example 3

The table contains the placement test scores and final scores of 20 students. To be eligible to take this course, the minimum placement test score required is 35.

a) Draw a scatter plot.

b) Determine the equation to predict the final score based on the placement test score.

c) Draw the regression line on the scatter plot.

d) If 60 is the passing grade for this course, below what placement test score would a student be denied entry to the course?

Placement Test	Course Grade
50	53
35	41
35	61
40	56
55	68
65	36
35	11
60	70
90	79
35	59
90	54
80	91
60	48
60	71
60	71
40	47
55	53
50	68
65	57
50	79